

Pranshu Patel

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Summary

Backend-focused Java developer with hands-on experience building secure Spring Boot APIs with JWT authentication, concurrency control, Redis locking, and automated testing. Built and deployed production-style backend systems using PostgreSQL, Docker, and RESTful architecture — with 90%+ test coverage and 300+ LeetCode problems solved.

Education

L.D. College of Engineering (LDCE)

Bachelor of Engineering in Information Technology — CGPA: 8.8/10

July 2023 – June 2027

Ahmedabad, Gujarat

Technical Skills

Languages: Java, SQL

Backend: Spring Boot, Spring Security, Hibernate, REST API, JDBC

Concepts: REST API Design, JWT Auth, RBAC, DTO Pattern, Pagination & Sorting, Transaction Management, Pessimistic Locking, Idempotency, Exception Handling, Input Validation

Testing & DevOps: JUnit 5, Mockito, Docker, Docker Compose, Git, Maven, Postman, Swagger/OpenAPI, Flyway, Resilience4j

Databases: MySQL, PostgreSQL, Redis

Projects

Secure Engineer Management API [\[Live Demo\]](#) — Spring Boot, Spring Security, PostgreSQL, Docker *Nov 2025 – Feb 2026*

- Built a Spring Boot REST API with **10+ secured endpoints** using layered architecture (Controller → Service → Repository); implemented **DTO layer** to eliminate sensitive field exposure and reduce payload size by **~30%**; added dynamic filtering via JPA Specifications with server-side **pagination & sorting**.
- Secured all endpoints with **JWT authentication** and **RBAC** via Spring Security; enforced ADMIN/USER access control at the method level using `@PreAuthorize` annotations.
- Achieved **90%+ service-layer test coverage** using JUnit 5 and Mockito with mocked dependencies; wrote controller-level integration tests with `MockMvc`.
- Implemented **centralized exception handling** using `@RestControllerAdvice` with custom HTTP error responses (400/401/403/404/409), enforced input validation via `@Valid` and `@NotBlank`, and added structured **SLF4J logging** across service, security, and exception layers.
- Documented all endpoints with **Swagger/OpenAPI**; containerized with **Docker** and deployed on Render using Neon PostgreSQL.

Ticket Booking System — Spring Boot, Spring Security, PostgreSQL, Redis, Docker, Flyway

Jan 2026 – Present

- Engineered a high-concurrency booking backend; validated correctness under load using an **8-thread, 1-seat concurrency test** (ExecutorService + CountDownLatch), confirming exactly **1 CONFIRMED booking** with 0 race conditions.
- Implemented **pessimistic locking** (PESSIMISTIC_WRITE) and **Redis soft locking** (SET NX EX) to prevent race conditions and hold seat selections before payment confirmation.
- Enforced **idempotency keys** on critical endpoints and managed schema evolution with **Flyway** migrations; orchestrated multi-container setup (API, PostgreSQL, Redis) with startup ordering and persistent volumes via **Docker Compose**.

Data Structures & Algorithms

300+ problems solved across Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Heaps, Graphs, Binary Search, Two Pointers, Sliding Window, Prefix Sum, Hashing, Greedy, Recursion & Backtracking, and Dynamic Programming.

Achievements

McKinsey Forward Program — Selected as a participant from a competitive applicant pool (Apr 2026 – Jun 2026)

School Topper — 1st in school, HSC Class 12 Science, scored 80.3% (GSEB, 2023)

Certifications

Introduction to Generative AI — Google (Badge) | **Java Certification Test** — KnowledgeGate